

Here is as detailed of a list as I can create with little time on what still needs to be done for my cerebellum project before we can publish our work. Firstly, I will give you access to our manuscript outline such that you know which figures we are trying to make where:

https://docs.google.com/document/d/1BIpa1BI_t5DU8-wL4zCSh6Zm78ucBAqkselo-u057eM/edit?usp=sharing

Next you will need participant information. If you don't have access to this excel spreadsheet, ask Minsu or HY for access:

[VML_MEG_Lab_Data.xlsx](#)

The participants that I am running for analysis are as follows. See the spreadsheet if you need their ages (11 children, 12 adults):

002	003	004	006	009	010	015	016	027	028	029
Child	Child	Child	Child	Child	Child	Child	Child	Child	Child	Child

011	012	013	014	018	019	021	022	025	026	030	031
Adult	Adult	Adult	Adult	Adult	Adult	Adult	Adult	Adult	Adult	Adult	Adult

Finally all of our data is stored on globus. Set up globus connect personal. Desktop9 is what works on my account, and it may work on yours for transferring files to/from globus, but you will have to check and work it out otherwise. It is worth setting up properly, since you will have to download and transfer many many folders of files. You will need access to globus, as Minsu or HY if you don't have access.

<https://app.globus.org/file-manager/collections?pane=2>

All of my work (codes and data) can be found in /Lab_Projects_backup/Max_Garson/ path from the main IMM Lab Data Storage in the file manager. **PLEASE READ THE READ_ME FILES** that are found in the scripts/pasd_data folders as you work. They are designed (like this is) to make things less confusing.

OK now that you are hopefully fully set, here are the tasks, *in order (important)* that we need to get done in order to publish our paper:

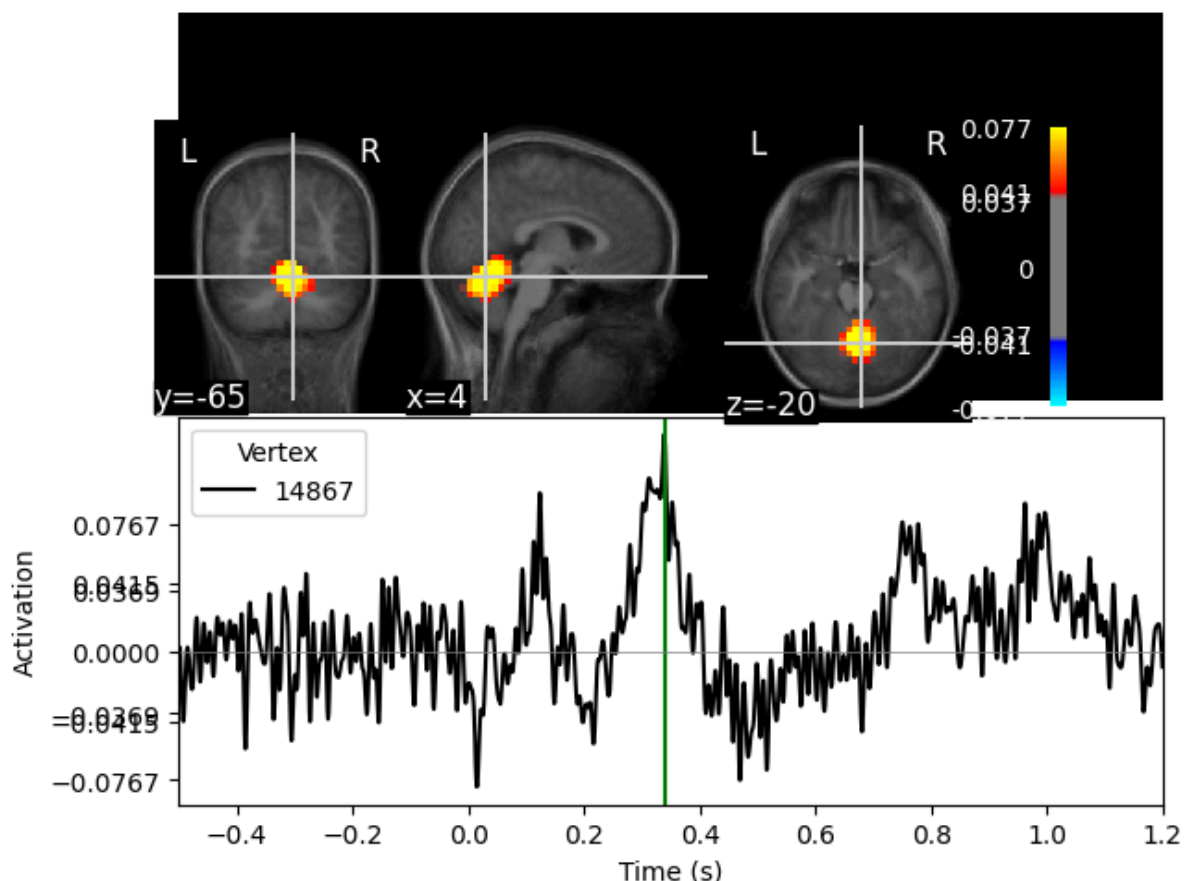
0.5) I will send you a video so that you can understand how the task was done/what it is. Familiarize yourself with what a PSD is and how it is used in neuroscience research. Learn some MEG lingo like epochs, beamforming. Learn where the cerebellum is.

- 1) We want to validate our PSD results (see PSD figures folder in the PSD data folder on globus). Because the brain produces strong post-movement beta activations in the motor cortices, we want to know whether we are capturing the beta signals in the power representations of the brain dynamics (PSDs). Our epochs are 1700ms long, and broken down roughly as follows:

- -500-0ms = time before stimulus shown to participant
- 0ms = stimulus appears
- 0-1200ms = participant response to stimulus
- Each epoch counts as one trial. The task is split into different learning blocks (baseline, early, late, washout, relearning) with ~30 trials each.

Because we only capture 1200ms of response time to the stimulus, if participants were to only stop moving their hand at ~1100ms, then we aren't keeping enough time to capture the post-movement beta activations, especially in the child group which takes longer to complete the task. Your job is then to go over the behavioural MEG data, and use the movement time code to extract one number e.g. 1300 ms that represents the average time before the child/adult groups stop their movement. Behavioural data is typically run in MATLAB and you can ask Minsu for some codes. Once you have the results, we can compare them with the figures in the PSD_figures folder in a meeting with HY and decide what the next best steps are for ensuring that our power analysis is done properly.

- 2) We want to make a figure using the Nilearn package (or something else like fsleyes, this is up to you and I haven't used either but would recommend nilearn) that is a *much* cleaner version of this:



Which shows where (in MNI coordinates) and when the cerebellum reaches its maximum activation. We want to spatially compare the adult/children group, meaning that the activations for each participant must be standardized to a common space, otherwise the same coordinates will designate different anatomy due to the head size mismatch between

children and adults. Currently all child brain activations have been morphed to my average child template, `vml_avg_child`, while the adults have been morphed to `fsaverage`. You must now re-morph everyone's STC plots to a new `fsaverage` template. Call `fetch_fsaverage` and name the MRI folder something else to avoid confusion. Then morph children and adults to this average subject. You will have to be careful when creating the average source-space for morphing. It must have the same structure as the current `stcs`, which is a surface for the cortical regions of the brain as usual, with the addition of a volume source space to model the cerebellum. We call this a mixed source space (surface + volume). The code for doing this is in `DeepSource_Localization_Using_Beamformer.ipynb`. See globus README. You will need to be vigilant and change the necessary to/from source spaces. Once everyone has been re-morphed, check the stc plots to make sure that they make sense. The activations should look very similar to the avg. STC plots that I have already generated, and it is important to compare them. Average together the child/adults separately across conditions for comparison. At this point you are ready to make the figure! Talk about slices/views with HY when you get here.

Recommendations

- 1) Don't overwrite any of my files. Create your own naming conventions, and whatever you make/redo just rename and call in as needed.
- 2) Write a new script for morphing, just copy/paste the code from my scripts that you need. That way you can see which files you need and it will be less messy.
- 3) DONT rename any of the participants. Stick to 'vml_meg_002' or 'vml_mri_002' etc.

